

# Francis Xiatian Zhang, PhD

Robot Vision · Medical Robotics · Computer Vision

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## Research Profile

Research Associate at the University of Edinburgh working on robot vision for medical robotics. Research combines geometric modelling and deep learning for robotic bronchoscopy, with an emphasis on depth estimation, visual odometry, airway segmentation, and reliable visual navigation.

## Research and Teaching Experience

- Nov 2024–present    **Research Associate · University of Edinburgh**  
Develops visual navigation and perception systems for robotic bronchoscopy, including airway segmentation, geometric graph construction, and failure diagnosis. PI: Dr Mohsen Khadem.
- Oct–Nov 2023;  
Jul–Sep 2024    **Research Assistant · Durham University**  
Contributed to the Edge Computing and Analytics 2.0 course, teaching model training, ONNX export, and deployment through Python APIs. PI: Dr Anish Jindal.
- Nov 2021–Jun 2024    **Demonstrator · Durham University**  
Supported laboratory teaching in Computational Thinking, Data Science, Programming for Data Science, and Text Mining.
- Apr–Jul 2022    **Research Assistant · Northumbria University**  
Developed multi-camera data collection and pose-based models for automatic assessment of CPR skills in nursing simulation. PI: Dr Merryn Constable.

## Education

- 2025    **PhD · Durham University**  
Geometric representations for clinical video analysis.
- MRes · King's College London**
- MSc · University of Southampton**
- Bachelor's degree · Beijing University of Chinese Medicine**

## Awards

- 2026    **ICRA Best Paper Award in Medical Robotics**  
Co-author of [Geometry-Aware Visual Odometry for Bronchoscopic Navigation via High-Gain Observer Fusion](#).
- 2020    **Dean's List Award for Outstanding Achievement**  
University of Southampton.

## Selected Publications

Selected work in robotics, computer vision, and biomedical engineering. Full publication list: [Google Scholar](#).

### 1. FDRAS: Failure Diagnosis and Repair for Airway Segmentation

**Francis Xiatian Zhang**, Kevin Dhaliwal and Mohsen Khadem

MICCAI 2026 · [paper](#) · [code](#)

### 2. Geometry-Aware Visual Odometry for Bronchoscopic Navigation via High-Gain Observer Fusion

Mohammadreza Kasaei, **Francis Xiatian Zhang**, Feng Li, Farshid Alambeigi, Kevin Dhaliwal and Mohsen Khadem

ICRA 2026 · [arXiv](#)

[Best Paper Award in Medical Robotics, ICRA 2026](#)

### 3. A Model-Based Framework for Assessing Operator Performance in Navigational Bronchoscopy

Zhaoxing Deng, David Hanley, **Francis Xiatian Zhang**, Kevin Dhaliwal and Mohsen Khadem

ICRA 2026 · [conference program](#)

### 4. ROOM: A Physics-Based Continuum Robot Simulator for Photorealistic Medical Datasets Generation

Salvatore Esposito, Matías Mattamala, Daniel Rebain, **Francis Xiatian Zhang**, Kevin Dhaliwal, Mohsen Khadem and Subramanian Ramamoorthy

ICRA 2026 · [arXiv](#)

### 5. Navigational Bronchoscopy in Critical Care via End-to-End Pose Regression

Emile Mackute, **Francis Xiatian Zhang**, Kevin Dhaliwal and Mohsen Khadem

MICCAI 2025 · [paper](#)

### 6. BREA-Depth: Bronchoscopy Realistic Airway-geometric Depth Estimation

**Francis Xiatian Zhang**, Emile Mackute, Mohammadreza Kasaei, Kevin Dhaliwal, Robert Thomson and Mohsen Khadem

MICCAI 2025 · [paper](#) · [arXiv](#) · [code](#)

### 7. Depth-Aware Endoscopic Video Inpainting

**Francis Xiatian Zhang**, Shuang Chen, Xianghua Xie and Hubert P. H. Shum

MICCAI 2024 · [paper](#) · [arXiv](#) · [code](#)

### 8. Pose-Based Tremor Classification for Parkinson's Disease Diagnosis from Video

Haozheng Zhang, Edmond S. L. Ho, **Francis Xiatian Zhang** and Hubert P. H. Shum

MICCAI 2022 · [paper](#) · [arXiv](#) · [code](#)

### 9. Cross-view Multimodal Vision-Based Assessment Framework for Traditional Chinese Medicine Rehabilitation Training

**Francis Xiatian Zhang**, Hao Yao, Shengxuan Chen, Hong Zhu, Hongxiao Jia, Sisi Zheng and Hubert P. H. Shum

IEEE TNSRE 2026 · [paper](#) · [arXiv](#) · [code](#)

### 10. Adaptive Anticipation: Adaptive Graph Learning From Spatial Information for Surgical Workflow Anticipation

**Francis Xiatian Zhang**, Jingjing Deng, Robert Lieck and Hubert P. H. Shum

IEEE TMRB 2024 · [paper](#) · [arXiv](#) · [code](#)

### 11. Pose-based Tremor Type and Level Analysis for Parkinson's Disease from Video

Haozheng Zhang, Edmond S. L. Ho, **Francis Xiatian Zhang**, Silvia Del Din and Hubert P. H. Shum

IJCARS 2024 · [paper](#) · [arXiv](#)